24/06/2026

Code A Phase 1

Aakash

Medical | IIT-JEE | Foundations

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MM : 720

Fortnightly Test for NEET-2027 (XII-Passed)_FT02A

Time : 180 Min.

Topics Covered:
Physics: Motion in a Plane, Laws of Motion
Chemistry: Structure of Atom, Classification of Elements and Periodicity in Properties
Botany: Cell Cycle & Cell Division, The Living World
Zoology: Biomolecules-II: (Proteins, types & functions, Lipids, Nucleic acids, Enzymes, Cofactors), Breathing & Exchange of Gases-I: (Upto mechanism of breathing)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

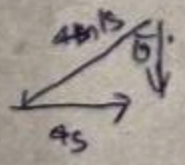
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PHYSICS

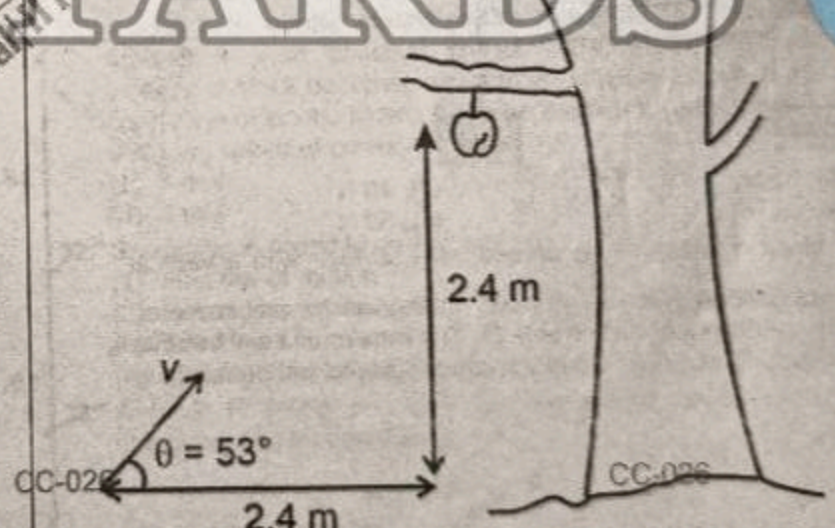
1. From the top of a tower of height 40 m, a ball is projected with a speed of 20 ms^{-1} at an angle of elevation 30°. The ratio of time taken by the ball to come back to same level and time taken to hit the ground is (g = 10 ms^{-2})
(1) 1 : 2 (2) 3 : 1
(3) 4 : 1 (4) 2 : 3

2. A person walking at 4 m/s, finds rain drops falling on his head vertically with speed of 4 m/s. The actual speed of rain drop is
(1) 6 m/s (2) 4 m/s
(3) $4\sqrt{2}$ m/s (4) 10 m/s

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3. A particle is to be projected from distance 2.4 m from base of a tree to hit an apple hanging at height 2.4 m. If angle of projection is fixed at 53°, then what should be velocity of projection? (g = 10 m/s^2)



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(1) 25 m/s (2) 20 m/s
(3) 15 m/s (4) 10 m/s

4. A particle is moving towards west with speed 40 ms^{-1} and then it turns towards north with speed 30 ms^{-1}. The direction of average acceleration for the concerned motion is
(1) 37° north of east (2) 53° north of east
(3) 30° south of west (4) 37° north of west

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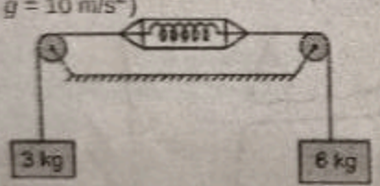
5. If X and Y components of a vector $\vec{P}$ have numerical values 3 and 4 respectively and X and Y component of $\left(\vec{P}+\vec{Q}\right)$ have numerical values 8 and 7 respectively, then the magnitude of $\vec{Q}$ is (consider all the vectors in first quadrant)
(1) 3 (2) 5
(3) $\sqrt{13}$ (4) $\sqrt{34}$

6. A set of three vectors are given as $\vec{A}=2\hat{j}+3\hat{j}+4\hat{k}, \vec{B}=5\hat{j}$ and $\vec{C}=2\hat{j}+6\hat{j}$ Then Addition of magnitudes of unit-vectors of $\vec{A}, \vec{B}$ and $\vec{C}$ would be
(1) 2 (2) 4
(3) 1 (4) 3

7. A projectile is given an initial velocity of $(2\hat{i}+3\hat{j})$m/s, where $\hat{i}$ is along horizontal and $\hat{j}$ is along the vertical. The equation of trajectory of projectile is (g = 10 m/s^2)
(1) $4y=3x^2-8x$ (2) $y=12x+10x^2$
(3) $8y-10x^2+12x=0$ (4) $10x^2-12x+8y=0$

8. **Assertion (A):** A body moving with uniform velocity is said to be in equilibrium.
Reason (R): For a body in equilibrium, no net external force should be acting on the body.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

9. In the given diagram, reading of spring balance will be (take g = 10 m/s^2)



(1) 30 N (2) 40 N
(3) 60 N (4) 80 N

10. A block of mass m_0 is placed on a surface with a vertical cross-section given by $y = 3x^2$ here y and x are in meter. The coefficient of friction is 0.75. The maximum height above the ground at which the block can be placed without slipping is
(1) $\frac{1}{m}$ m (2) $\frac{m}{1}$ m
(3) $\frac{3}{m}$ m (4) $\frac{1}{3}$ m

11. If impulse I varies with time t as I(kg m/s) = $(10t^2 - 4t)$. The change in momentum is zero in time t = 0 to
(1) $t=\frac{1}{2}$ s (2) $t=\frac{1}{4}$ s
(3) $t=\frac{2}{5}$ s (4) $t=\frac{1}{10}$ s

12. **Statement A:** The force of friction only acts between the two surfaces when one surface is in relative motion w.r.t. the other.
Statement B: The phenomenon of raising outer edge of a curved road above the inner edge is called banking of the road.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Statement A is correct and statement B is incorrect
(4) Statement A is incorrect and statement B is correct

13. A coin placed on a rotating turntable just slips, if it is placed at a distance of 4 cm from the centre. If the angular velocity of the turntable is doubled, it will just slip at a distance of
(1) 4 cm (2) 2 cm
(3) 1 cm (4) 0.5 cm

14. A person wants to reach exactly opposite point on the other bank of a 1 km wide river. If he swims with speed 4 m/s at an angle 120° with direction of flow of water. The rate of flow of river is
(1) 2 m/s (2) 1 m/s
(3) 2.5 m/s (4) 1.5 m/s

15. Rain is falling vertically with a speed of 20 ms^{-1}. A girl rides a bicycle with a speed of 10 m/s in west direction. The direction θ in which she holds her umbrella to save herself from rain is
(1) $\theta = \tan^{-1}(2)$ from vertical towards west
(2) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ from horizontal towards west
(3) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ from vertical towards east
(4) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ from vertical towards west

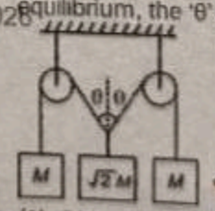
16. A block of mass 2 kg is to be kept at rest against a rough vertical wall (μ = 0.4) by applying a horizontal force F. The minimum value of F is (g = 10 m/s^2)
(1) 20 N (2) 40 N
(3) 50 N (4) 45 N

17. In air, the relative acceleration of one projectile with respect to another projectile is [g is acceleration due to gravity]
(1) g (2) $\frac{g}{2}$
(3) 2 g (4) Zero

18. A block of mass 6 kg is at rest on a rough inclined surface of angle of inclination 30° and $\mu=\frac{\sqrt{3}}{2}$ The force of friction acting on block is (g = 10 m/s^2)
(1) 45 N (2) 30 N
(3) 15 N (4) $30\sqrt{3}$

19. The vector sum of the forces 10 N and 6 N can be
(1) 2 N (2) 8 N
(3) 18 N (4) 20 N

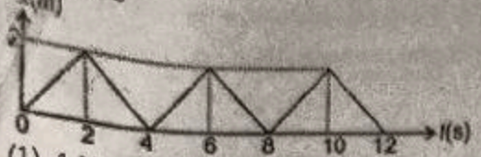
20. In the given arrangement, for the system to remains in equilibrium, the 'θ' should be



(1) 30° (2) 60°
(3) 45° (4) 0°

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The position(x)-time(t) graph of one dimensional motion of a body of mass 0.5 kg is shown. The magnitude of each impulse is

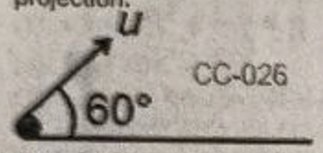


(1) 1 N s
(2) 2 N s
(3) 0.5 N s
(4) 1.5 N s

22. An object A is moving with constant velocity $\vec{v}_A = (5\hat{i} - 2\hat{j})$ m/s and object B is moving with constant velocity $\vec{v}_B = (3\hat{i} + 5\hat{j})$ m/s.. The velocity of B w.r.t. A is

(1) $(2\hat{i} - 7\hat{j})$ m/s
(2) $(-2\hat{i} + 7\hat{j})$ m/s
(3) $-(2\hat{i} + 7\hat{j})$ m/s
(4) $(2\hat{i} + 7\hat{j})$ m/s

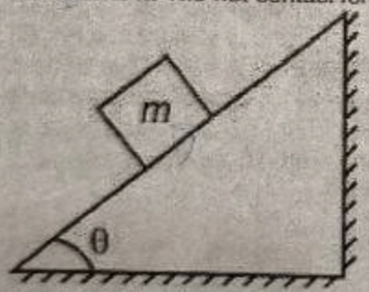
23. An object is projected with speed u at an angle 60° with horizontal. Find the ratio of magnitude of tangential acceleration to centripetal acceleration at point of projection.



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(1) $\sqrt{6}$
(2) $\sqrt{5}$
(3) $\sqrt{3}$
(4) $\sqrt{2}$

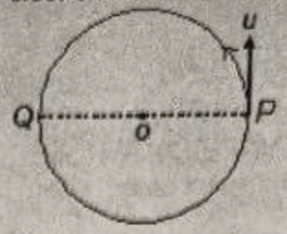
24. Given below are two statements:
Statement-I: A block is at rest on an inclined plane as shown. If the coefficient of friction between the block and inclined plane is μ, then frictional force acting on block is $mg\sin\theta$.
Statement-II: The net contact force on the block is mg.



In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

25. Consider the following statements:
(A) Impulse is equal to change in momentum.
(B) Action and reaction forces can cancel each other.
(C) A light string has the same tension throughout.
(D) Non-accelerated frame of reference is called inertial frame.
The correct statement(s) is/are
(1) Only (A)
(2) Both (B) and (C)
(3) (A), (B) and (C)
(4) (A), (C) and (D)

26. Let an electron revolve around nucleus with a constant speed u. The magnitude of its change in velocity when electron move from P to Q is



(1) u
(2) $2u$
(3) Zero
(4) $\sqrt{2}u$

27. **Assertion (A) :** Pulling a body on a rough surface is easier than pushing it to slide.
Reason (R) : Pushing increases the normal reaction and hence the force of friction.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

28. To avoid slipping while walking on ice, one should take smaller steps because of the
(1) Friction coefficient of ice is small
(2) Friction coefficient of ice is large
(3) Large normal reaction
(4) Small normal reaction

29. If the resultant of two vectors of magnitudes A and $2A$ is perpendicular to the vector having magnitude A, then the angle between the vectors is
(1) 60°
(2) 90°
(3) 120°
(4) 135°

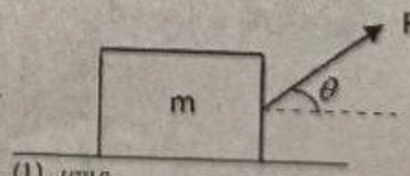
30. If a conical pendulum of length (L) makes an angle (θ) with the vertical then time period (T) is related with θ as
(1) $T^2 \propto \cos\theta$
(2) $T^2 \propto \sin\theta$
(3) $T^2 \propto \cos^2\theta$
(4) $T^2 \propto \sin^2\theta$

31. A block is gently placed on a conveyer belt moving horizontally with constant speed. After 3 seconds, the velocity of block becomes equal to the velocity of belt. If the coefficient of kinetic friction between the block and the belt is 0.3, then speed of conveyer belt is
(1) 4 m/s
(2) 6 m/s
(3) 8 m/s
(4) 9 m/s

32. Swimming is possible on account of
(1) First law of motion
(2) Second law of motion
(3) Third law of motion
(4) Newton's law of gravitation

33. A force F acting on block as shown and block remain stationary find frictional force

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(1) μmg
(2) $F\cos\theta$
(3) $\frac{\mu mg}{\sin\theta + \mu\cos\theta}$
(4) $\frac{\mu mg}{\sqrt{1+\mu^2}}$

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34. A rocket consumes 50 kg fuel per second. The exhaust speed of a gases w.r.t rocket is 3 kms^{-1}. The force exerted on the rocket by the fuel is

(1) 15×10^5 N (2) 15×10^4 N
(3) 1.5×10^4 N (4) 2.5×10^3 N

35. A cricket ball of mass 250 g moving with a speed of 10 m/s is hit back by a bat and returns with the same speed within 0.01 s. The force exerted by the bat is

(1) 200 N (2) 400 N
(3) 500 N (4) 250 N

36. A particle has an initial velocity of $\left(6\hat{i}+10\hat{j}\right)$ m s^{-1} and acceleration of $\left(2\hat{i}-3\hat{j}\right)$ m s^{-2}. Its speed at 2 s is

(1) $10\sqrt{2}$ m s^{-1} (2) $20\sqrt{2}$ m s^{-1}
(3) $\sqrt{116}$ m s^{-1} (4) $\sqrt{104}$ m s^{-1}

37. On a mine site, a rock is exploded. On explosion, rock breaks into three parts. Two parts go off at right angles to each other. These two parts are 1 kg and 2 kg moving with velocity of 12 m/s and 8 m/s respectively. If third part has mass of 5 kg, its speed would be

(1) 6 m/s (2) 4 m/s
(3) 12 m/s (4) 5 m/s

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38. A particle projected from earth surface is moving in vertical plane (x-y plane) such that its trajectory is given by the equation, $y = x - \frac{x^2}{80}$, where x & y are in metre.

For this particle match column-I with column-II and tick the correct option.

	Column A		Column B
A.	Horizontal range (in m)	(P)	45
B.	Angle of projection with horizontal (degree)	(Q)	80
C.	Maximum height gained by particle (m)	(R)	20
D.	Speed of projection (m s^{-1})	(S)	$20\sqrt{2}$

(1) A(S), B(R), C(Q), D(P) (2) A(P), B(Q), C(S), D(R)
(3) A(Q), B(P), C(S), D(R) (4) A(Q), B(P), C(R), D(S)

39. The speed of a projectile u reduces by 50% on reaching the maximum height. The maximum height will be

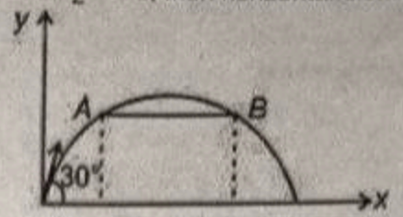
(1) $\frac{3u^2}{4g}$ (2) $\frac{3u^2}{8g}$
(3) $\frac{3u^2}{2g}$ (4) $\frac{u^2}{4g}$

40. An object of mass 2 kg starts from rest under a force $\vec{F} = 4t\hat{i} + 5\hat{j}$. The final speed of the object after 2 s will be (in m/s)

(1) $\sqrt{37}$ (2) $\sqrt{39}$
(3) $\sqrt{164}$ (4) $\sqrt{41}$

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41. A particle is thrown with a velocity u = 40 m/s. If the partic passes points A and B, as shown in figure, at time t_1 = 1 and t_2 = 3 s, then horizontal distance AB is equal to

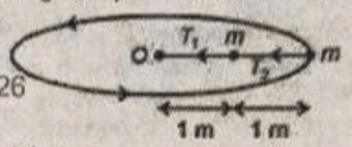


(1) 40 m (2) 20 m
(3) $40\sqrt{3}$ m (4) $20\sqrt{3}$ m

42. A particle moves in y-z plane according to the displacement equations $y = a_0 \sin 2\pi t$ and $z = a_0 \cos 2\pi t$. The trajectory of the particle will be

(1) An elliptical path (2) A circular path
(3) A parabolic path (4) A straight line path

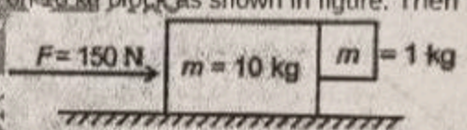
43. Two small spheres of equal masses are attached to a string of length 2 m, as shown in figure. The string and spheres are whirled in a horizontal circle about point O at constant angular speed. The ratio of tension $\frac{T_1}{T_2}$ is



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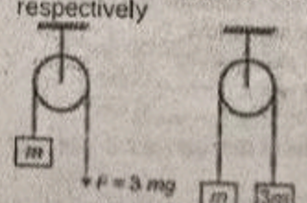
(1) 1 : 2 (2) 2 : 3
(3) 1 : 3 (4) 3 : 2

44. The coefficient of friction between 10 kg block and 1 kg block is μ = 0.8 and no friction is between 10 kg block and horizontal surface. If a horizontal force F = 150 N is applied on 10 kg block as shown in figure. Then



(1) 1 kg block will move upward
(2) 1 kg block will fall down
(3) 1 kg block will move along with 10 kg block without falling down or moving upward
(4) Acceleration of the block is 10 m/s^2

45. In the arrangement shown in two figures, the mass m will ascend with an acceleration, in figure (i) and figure (ii) respectively



CC-026 (i) (ii) CC-026

(1) $g, \frac{g}{3}$ (2) $2g, \frac{g}{2}$
(3) $\frac{g}{3}$ (4) $\frac{g}{2}, \frac{g}{2}$

CHEMISTRY

46. The ratio of "charge to mass (e/m)" would be maximum for

(1) Proton (2) Electron
(3) Neutron (4) Alpha particle

47. Which of the following pair of species would contain same number of neutrons?

(1) $^{13}_{6}C$ and $^{14}_{7}N$ (2) $^{16}_{8}O$ and $^{24}_{12}Mg$
(3) $^{59}_{26}Fe$ and $^{87}_{38}Sr$ (4) $^{35}_{17}Cl$ and $^{40}_{19}Ca$

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oose the incorrect statement about canal rays.
(1) They possess positive charge
(2) They travel in a straight line in the absence of electrical or magnetic field
(3) They emerge from anode and move towards the cathode
(4) Properties of canal rays depend upon the nature of the gas present in cathode ray tube

49. According to the Rutherford's nuclear model of atom
(a) Electrons and the nucleus are held together by electrostatic forces of attraction
(b) The positive charge and most of the mass of the atom was densely concentrated in extremely small region called nucleus
(c) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called orbits
The correct statement(s) is/are
(1) (b) and (c) only (2) (a), (b) and (c)
(3) (a) and (c) only (4) (a) only

50. The frequency of red radiation having wavelength 700 nm approximately is
(1) $4.3 \times 10^{14}\ s^{-1}$ (2) $7.9 \times 10^{14}\ s^{-1}$
(3) $5.7 \times 10^{14}\ s^{-1}$ (4) $2.5 \times 10^{14}\ s^{-1}$

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51. Consider the following statements
(a) Frequency of emitted radiation of maximum intensity from a black body goes from a lower frequency to a higher frequency as the temperature increases
(b) Photoelectric effect and diffraction phenomenon can be explained by wave nature of electromagnetic radiation.
(c) Rutherford's α-particle scattering experiment led to the discovery of nucleus.
(d) Work function (W_0) for Li is higher than Na.
The correct statement(s) is/are
(1) (a), (c) and (d) only (2) (a) only
(3) (b) and (c) only (4) (b), (c) and (d) only

52. A 10 W bulb emits 400 nm radiation, the rate of emission of photons per second is
($h = 6.626 \times 10^{-34}$ Js)
(1) 2×10^{12} (2) 5×10^{15}
(3) 5×10^{17} (4) 2×10^{19}

53. A gas absorbs a photon of 600 nm and then re-emits two photons. If one of the photon has a wavelength 900 nm, then the wavelength of other photon will be
(1) 900 nm (2) 450 nm
(3) 1800 nm (4) 600 nm

54. Match List I with List II.

	List I (Quantum number)		List II (Information provided)
a.	n	(i)	Shape of orbital
b.	l	(ii)	Orientation of orbital
c.	m_l	(iii)	Orientation of the spin of electron
d.	m_s	(iv)	Size of orbital

Choose the correct answer from the option given below
(1) a(i), b(iv), c(ii), d(iii) (2) a(iv), b(i), c(ii), d(iii)
(3) a(i), b(ii), c(iv), d(iii) (4) a(iv), b(i), c(iii), d(ii)

55. The shortest wavelength of H-atom in the Balmer series is λ_1. The longest wavelength in the Paschen series of Li^{2+} ion is
(1) $\frac{4\lambda_1}{7}$ (2) $\frac{7\lambda_1}{4}$
(3) $\frac{7}{4\lambda_1}$ (4) $\frac{4}{7\lambda_1}$

56. Consider the following statements about Bohr's Model of an atom
(i). Angular momentum of an electron in the 3rd orbit of hydrogen atom is $\frac{3h}{\pi}$
(ii). The radius of first stationary state of hydrogen is 52.9 pm.
(iii). The total energy of an electron in second stationary state for hydrogen atom is -0.545×10^{-18} J.
The correct statement(s) is/are
(1) (i), (ii) & (iii) (2) (ii) only
(3) (ii) & (iii) only (4) (iii) only

57. Based on the Bohr's model match List I with the list II

	List I		List II
a.	Kinetic energy of the electron in the nth orbit	(i)	$\propto n$
b.	Angular momentum of the electron in the nth orbit	(ii)	$\propto n^2$
c.	Radius of nth orbit	(iii)	$\propto n^{-1}$
d.	Velocity of electron in the nth orbit	(iv)	$\propto n^{-2}$

Choose the correct option
(1) a(iv), b(i), c(iii), d(iii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iv), b(iii), c(ii), d(i)

58. A football has a mass of 50g and a speed of 30 m/s. If the speed can be measured within accuracy of 2%.
Calculate uncertainty in the position (in meters)
(1) 1.63×10^{-33} (2) 1.79×10^{-31}
(3) 1.72×10^{-32} (4) 1.75×10^{-33}

59. Match List-I with List-II.

List-I			List-II	
	(Lower energy Level) n_1	(Higher energy Level) n_2		Series
(a)	1	2, 3...	(i)	Brackett
(b)	4	5, 6...	(ii)	Paschen
(c)	3	4, 5...	(iii)	Pfund
(d)	5	6, 7...	(iv)	Lyman

The **correct** match is
(1) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
(2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

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60. Which among the following is the incorrect set of quantum numbers

	n	l	m	s
(1)	4	2	−1	$+\frac{1}{2}$
(2)	5	1	1	$-\frac{1}{2}$
(3)	2	0	0	$+\frac{1}{2}$
(4)	3	2	3	$-\frac{1}{2}$

(1) (1) (2) (2)
(3) (3) (4) (4)

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61. The radius ratio of 3rd Bohr orbit of He^+ ion to 2nd Bohr orbit of Li^{2+} ion is
(1) 8 : 27 (2) 27 : 8
(3) 4 : 9 (4) 9 : 4

62. Wavelength observed in visible region of hydrogen spectrum is expressed as $\lambda = \frac{Kn^2}{n^2-4}$. The value of K in terms of Rydberg constant (R) is
(1) $\frac{3R}{4}$ (2) $\frac{4}{R}$
(3) 3R (4) $\frac{R}{7}$

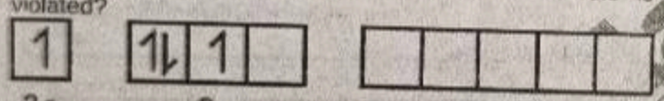
63. Which of the following statements is incorrect?
(1) 1s electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon.
(2) The radius of an atomic nucleus is of the order of 10^{-15} m.
(3) Splitting of spectral lines in the presence of electrical field is called Zeeman effect
(4) Rutherford's experiment, which established the nuclear model of the atom, used a beam of helium nuclei, which impinged on a metal foil and got scattered.

64. Match List-I with List-II and choose the correct code.

	List-I (Orbitals)		List-II (Number of radial nodes)
(a)	2p	(i)	1
(b)	3p	(ii)	3
(c)	3s	(iii)	0
(d)	4s	(iv)	2

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

65. In the given configuration representation which rule is violated?



(A) Aufbau principle
(B) Pauli's exclusion principle
(C) Hund's rule of maximum multiplicity
Choose the correct option.
(1) (A), (B) and (C) (2) (A) and (C) only
(3) (B) and (C) only (4) (B) only

66. The lowest value of n (Principal quantum number) that allows g orbitals to exist is
(1) 5 (2) 6
(3) 4 (4) 3

67. If KE(Kinetic energy) of an electron is increased 4 times, the wavelength of de-Broglie wave associated with it would become
(1) 4 times (2) 2 times
(3) $\frac{1}{2}$ times (4) $\frac{1}{4}$ times

68. Number of unpaired electrons present in Co^{2+} ion is
(1) 3 (2) 4
(3) 2 (4) 5

69. Number of electrons in Fe^{3+} ion having m = 0 is
(1) 6 (2) 5
(3) 11 (4) 20

70. Exchange energy of an electron is the energy released when it exchanges its position with
(1) The electron having opposite spin present in non-degenerate orbitals
(2) The electron having same spin present in degenerate orbitals
(3) The electron having same spin present in non-degenerate orbitals
(4) The electron having opposite spin present in degenerate orbitals

71. Given below are two statements.
Statement I: In Mendeleev's periodic table, the properties of the elements are periodic function of their atomic weights.
Statements II: In modern periodic table, the physical and chemical properties of the elements are periodic functions of their atomic numbers.
In light of the above statements, choose the correct answer.
(1) Statement I is correct but statement-II is incorrect
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct.

72. Match the following columns I and II.

	Column I (Element)		Column II (Number of unpaired electron(s))
a.	Copper	(i)	5
b.	Manganese	(ii)	4
c.	Chromium	(iii)	1
d.	Iron	(iv)	6

Choose the correct option.
(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

73. Eka - aluminium and Eka - silicon in Mendeleev's periodic table respectively are
(1) Ga, Ta (2) Ga, Sn
(3) Ge, Ga (4) Ga, Ge

74. Dobereiner's Triads among the following are
(a) Li, Na, K
(b) Ca, Sr, Ba
(c) C, N, O
(d) Cl, Br, I
(1) (a) and (b) only (2) (b), (c) and (d) only
(3) (a), (b) and (d) only (4) (a) and (c) only

75. Match List-I with List-II

List-I(Atomic Number)	List-II (IUPAC official Name)
(a) 108	(i) Seaborgium
(b) 102	(ii) Rutherfordium
(c) 104	(iii) Nobelium
(d) 106	(iv) Hassium

Choose the correct match
(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

76. In the modern periodic table, the element with atomic number 33 belongs to
(1) 5th period and 15th group
(2) 3rd period and 13th group
(3) 4th period and 13th group
(4) 4th period and 15th group

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ment with Z = 60 belongs to

	Block	Period	Group
(1)	f	7th	3rd
(2)	d	6th	5th
(3)	f	6th	3rd
(4)	d	5th	5th

(1) (1)
(3) (3)
(2) (2)
(4) (4)

78. The element which is not a transuranium element is
(1) Plutonium
(2) Curium
(3) Thorium
(4) Berkelium

79. Which group elements are known as chalcogens?
(1) 13th group
(2) 14th group
(3) 15th group
(4) 16th group

80. Correct order of ionic radii is
(1) $O^{2-} > N^{3-} > Mg^{2+} > Na^{+}$
(2) $Mg^{2+} > Na^{+} > O^{2-} > N^{3-}$
(3) $N^{3-} > O^{2-} > Na^{+} > Mg^{2+}$
(4) $Na^{+} > Mg^{2+} > O^{2-} > N^{3-}$

81. Which of the following pair have nearly the same atomic radii?
(1) Zn, Mo
(2) Co, Ni
(3) Zn, Cd
(4) Ti, Hf

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82. Formation of $O^{2-}(g)$ occurs as follows, comment on the nature of two steps
(i) $O(g) + e^{-} \longrightarrow O^{-}(g)$
(ii) $O^{-}(g) + e^{-} \longrightarrow O^{2-}(g)$
(1) Step (i) is exothermic and step (ii) is endothermic
(2) Both the steps are exothermic
(3) Both the steps are endothermic
(4) Step (i) is endothermic and step (ii) is exothermic

83. Correct order of first ionization potential among the elements Be, B, C, N, O is
(1) B < Be < C < N < O
(2) B < Be < C < O < N
(3) Be < B < C < O < N
(4) Be < B < N < O < C

84. The correct order of negative electron gain enthalpy is
(1) Se < S < Br < F
(2) Se < F < S < Br
(3) F < Br < Se < S
(4) F < Se < S < Br

85. The element with highest negative electron gain enthalpy will belong to
(1) Group 17, period 3
(2) Group 17, period 2
(3) Group 18, period 2
(4) Group 16, period 2

86. Match the List-I with List-II.

List I	List II
(a) Nickel	(i) Non-metal
(b) Chlorine	(ii) Lanthanoid
(c) Germanium	(iii) Transition metal
(d) Samarium	(iv) Metalloid

Choose the correct match.
(1) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
(3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

87. Identify the correct order of electronegativity for the given species: (on Pauling scale)
(1) Si < C < N < O
(2) Si < N < C < O
(3) C < Si < O < N
(4) C < N < Si < O

88. Electronegativity of two elements A and B are 4 and 3 respectively. The percentage ionic character in A – B bond according to Hannay and Smith equation is
(1) 25%
(2) 30%
(3) 16.5%
(4) 19.5%

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89. Match List-I with List-II.

List-I	List-II
(a) NO	(I) Acidic
(b) CO_2	(II) Basic
(c) ZnO	(III) Neutral
(d) CaO	(IV) Amphoteric

Choose the correct answer from the options given below.
(1) (a)(II), (b)(IV), (c)(I), (d)(III)
(2) (a)(III), (b)(I), (c)(II), (d)(IV)
(3) (a)(III), (b)(I), (c)(IV), (d)(II)
(4) (a)(I), (b)(III), (c)(IV), (d)(II)

90. The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ respectively are
(1) +2 and 5
(2) +3 and 3
(3) +3 and 5
(4) +3 and 6

BOTANY

91. A bacterium divides amitotically in 20 minutes. A suspension culture of these bacteria containing 10^3 cells/ml is grown for 160 minutes. Find out cell concentration per ml in 160 minutes.
(1) 8×10^3
(2) 10^{768}
(3) 256×10^3
(4) 160×10^3

92. Syncytium formation is due to
(1) Failure of karyokinesis after cytokinesis
(2) Failure of cytokinesis after karyokinesis
(3) Failure of both karyokinesis and cytokinesis
(4) Occurrence of cytokinesis after karyokinesis

93. Select the **correct** sequence of events taking place during mitotic anaphase.
A – Chromatids separate
B – Centromeres split
C – Chromatids move to opposite poles
(1) B → A → C
(2) C → B → A
(3) A → B → C
(4) A → C → B

94. Given are two statements Assertion (A) and Reason (R).
Assertion (A): DNA replication does not take place just before meiosis II.
Reason (R): DNA is already present in double amount in each chromosome as compared to that of G_1 phase.
In the light of above statements select the **correct** option.
(1) Both (A) and (R) are false
(2) Both (A) and (R) are true and (R) is correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

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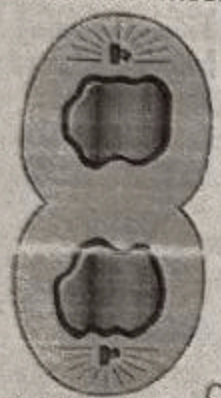
95. Gap phase 1 of interphase does not involve synthesis of
(1) Nucleotides
(2) Amino acids
(3) RNA
(4) DNA

96. Duration of cell cycle for a yeast cell is
(1) 90 hours
(2) 24 minutes
(3) 90 minutes
(4) 24 hours

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97. Select the events that are associated with the interphase.
a. Duplication of centrioles.
b. Attachment of spindle fibres to the kinetochores.
c. Exchange of genetic material between homologous chromosomes.
d. Synthesis of tubulin proteins required for forming mitotic apparatus.
e. Chromatin condensation.
The **correct** ones are
(1) a, c, d and e only (2) a and d only
(3) b and e only (4) c, d and e only

98. Consider the given figure regarding the cell division and choose the **incorrect** statement for it.



(1) Nucleolus, Golgi bodies and ER reappear in the daughter cell.
(2) This phase marks the initiation of nuclear division
(3) Mitotic spindle disappears
(4) Chromosomes are clustered at opposite poles and lose their identity

99. Interphase and interkinesis differs from each other as:
a. Former lasts more than 95% of the duration of cell cycle.
b. Latter occurs in the cells that have twice the amount of DNA than parent cell.
c. Former is a preparatory phase for upcoming cell division.
The **correct** one(s) is/are
(1) Only a (2) Only c
(3) Only a and b (4) All a, b and c

100. Double metaphase plate is formed during
(1) Prophase II (2) Metaphase I
(3) Anaphase II (4) Metaphase II

101. If there are 40 chromosomes in G_1-phase, what will be the number of bivalents in zygotene stage?
(1) 30 (2) 80
(3) 10 (4) 20

102. Phragmoplast during cell plate formation is formed by
(1) Mitochondria (2) Golgi complex
(3) Lysosome (4) Chloroplast

103. The metaphase plate is
(1) Formed by the endoplasmic reticulum
(2) Expanded from one spindle pole to the other
(3) Composed of microtubules and plasma membrane
(4) An imaginary plane where the centromere of chromosomes align themselves

104. The most dramatic period of the cell cycle is
(1) G_1 phase (2) M phase
(3) G_2 phase (4) S phase

105. Spireme stage of chromosomes is associated with
(1) Early prophase (2) Late prophase
(3) Metaphase (4) Telophase

106. Which of the given statement is **not** related to cytokinesis in a plant cell?
(1) New cell wall formation begins with the formation of a precursor called cell-plate
(2) Distribution of plastids in two daughter cells
(3) Wall formation starts in the centre
(4) Formation of a furrow in the plasma membrane

107. If a cell has 14 chromosomes and 2C DNA in G_1 phase. What will be the number of chromosomes and DNA amount respectively, after G_2 phase?
(1) 14, 2C (2) 7, 4C
(3) 14, 4C (4) 7, 2C

108. Resting phase of cell cycle is
(1) Prophase (2) Interphase
(3) M-phase (4) Only S-phase

109. Mitosis generally results in
(1) Introduction of variation in the genetic material.
(2) Disturbance in the ratio between the nucleus and the cytoplasm by increasing the cytoplasmic content.
(3) The production of haploid daughter cells with identical genetic complement from diploid mother cells.
(4) Restoring the surface area to volume ratio of the cell.

110. Read the following statements and select the **correct** option.
Statement A: Cell cycle is the period between the formation of new daughter cells and their further division.
Statement B: Interphase is a biosynthetic phase in which cell duplicates its organelles and replicates its DNA.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

111. Mark the following statements as true **(T)** or false **(F)** and select the **correct** option.
A. Morphology of chromosomes is best studied during metaphase.
B. Nucleolus reappears during prophase whereas disappears during telophase.
C. All organisms form gametes through meiosis as gametes are always haploid.

	A	B	C
(1)	T	T	T
(2)	T	F	F
(3)	F	F	T
(4)	T	F	T

(1) (1) (2) (2)
(3) (3) (4) (4)

112. Which of the following is **not** a significance of meiosis?
(1) Formation of gametes
(2) Increases genetic variability
(3) Reduction of chromosome number to half
(4) Replaces old, dead and worn out cells

113. Select the events which occur twice during meiosis.
a. Nuclear division
b. Division of cytoplasm
c. DNA replication
d. Bivalent formation
e. Equational division
(1) a and b only (2) a, b and e
(3) a, b and d (4) c, d and e

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Select the **correct** statement w. r. t. anaphase I.
(1) At the onset, each chromosome arranged at the metaphase plate is split into chromatids simultaneously
(2) The homologous chromosomes separate, while sister chromatids remain associated at their centromeres.
(3) Chromosomes undergo some dispersion and completely decondense on the poles.
(4) Degradation of interzonal fibres can be observed.

115. In final stage of prophase-I of meiosis
(1) Chiasmata formation occurs
(2) Crossing over occurs
(3) Recombinase enzyme gets fully activated
(4) Terminalisation of chiasmata takes place

116. How many mitotic divisions are required to produce 256 cells?
(1) 8 (2) 7
(3) 127 (4) 255

117. Activity of recombinase enzyme is seen at
(1) Leptotene (2) Pachytene
(3) Diplotene (4) Diakinesis

118. A tetrad
(1) Is a pair of homologous chromosomes
(2) Is clearly visible in zygotene stage
(3) Consists of two chromatids
(4) Consists of four centromeres

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119. Observe the following events:
a. Formation of the site of active ribosomal RNA synthesis.
b. Formation of the cell plate in the centre to form two daughter cells.
c. Complete disappearance of perinuclear space.
d. Centrosome moving towards the pole.
e. Attachment of spindle fibres to the sites found on the primary constriction.
f. Splitting of primary constriction.
Select the option depicting the **correct** sequence of events that occur during the formation of liquid endosperm in coconut.
(1) d → c → e → f → a → b (2) c → e → f → d → b
(3) e → f → d → c → a (4) c → e → f → a

120. In oocytes of some vertebrates, which of the following stage in meiosis can lasts for years?
(1) Leptotene (2) Zygotene
(3) Pachytene (4) Diplotene

121. Choose the **incorrect** match from the following:
(1) G_0 phase - Metabolically active
(2) Interzonal fibres - Dissolve during anaphase
(3) Metaphase chromosomes - Thickest and shortest
(4) Early prophase - Ends of chromatids are not visible

122. Read the given statement.
"Rice is a monocot and all monocots are plants"
In the above statement – rice, monocot and plants are
(1) Same taxa at same rank
(2) Different taxa at same rank
(3) Same taxa at different rank
(4) Different taxa at different rank

123. State true (T) or false (F) and select the **correct** option
A. As we go higher from species to kingdom the number of common characteristics goes on decreasing.
B. Tiger, leopard and lion are put along with dogs in the same family.
C. The highest common taxonomic category among mango and wheat is class.
D. Potato and brinjal are two different species but both belong to family Anacardiaceae.
(1) A(T), B(T), C(F), D(T) (2) A(T), B(F), C(T), D(F)
(3) A(F), B(F), C(F), D(F) (4) A(T), B(F), C(F), D(F)

124. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Taxonomy	(i)	These are distinct biological entities and not merely morphological aggregates
b.	Systematics	(ii)	A process of classification, based on characteristics, all living organisms can be classified into different taxa
c.	Classification	(iii)	It takes into account evolutionary relationships between organisms
d.	Taxonomic categories	(iv)	Process by which anything is grouped into convenient categories based on some easily observable characters

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(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(ii), b(i), c(iv), d(iii) (4) a(iv), b(iii), c(i), d(ii)

125. Choose the **correct** statements w.r.t. binomial nomenclature.
A. Name of author appears after specific epithet
B. Names are generally in Latin language
C. When handwritten, both generic and specific names are underlined together
D. Genus name starts with small letter and specific epithet starts with capital letter
(1) A and B (2) B and C
(3) C and D (4) A and D

126. The immediate next lower category to kingdom in taxonomic hierarchy for plants is
(1) Genus (2) Division
(3) Class (4) Order

127. Family is placed between
(1) Order and genus (2) Class and order
(3) Phylum and class (4) Genus and species

128. Which evolutionary biologist is famously known as the "Darwin of the 20th century"?
(1) Ernst Mayr
(2) Antonie Von Leeuwenhoek
(3) Rudolf Virchow
(4) G.N Ramachandran

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129. Biodiversity is referred as
(1) Diverse environment of a particular species
(2) The number and types of organisms on earth
(3) Availability of different food resources for a specific organism
(4) Variation among the offspring of an organism

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130. Which of the following taxa belong to the same rank?
(1) *Homo, Felis, pardus*
(2) Solanaceae, *leo*, Poaceae
(3) *Mangifera, Solanum, Panthera*
(4) Hominidae, *Triticum*, Dicotyledonae

131. Carnivora includes
(1) Group of organisms belonging to related orders
(2) Group of organisms belonging to related classes
(3) Group of organisms belonging to related families
(4) Group of organisms which are similar in all features

132. Choose the **odd** one w.r.t. class.
(1) Mammalia
(2) Monocotyledonae
(3) Arthropoda
(4) Dicotyledonae

133. Which of the following is **incorrect** w.r.t. species?
(1) A group of individual organisms with fundamental similarities
(2) Two different species breed together to produce fertile offsprings
(3) Human beings belong to the species *sapiens*
(4) *Panthera* has many specific epithet as *tigris, leo* and *pardus*

134. ICBN stands for
(1) Indian Code of Botanical Nomenclature
(2) International Congress of Biological Names
(3) Indian Criteria for Botanical Nomenclature
(4) International Code for Botanical Nomenclature

135. *Datura* is placed in the family
(1) Poaceae
(2) Anacardiaceae
(3) Solanaceae
(4) Muscidae

ZOOLOGY

136. Complete the analogy by selecting the **correct** option w.r.t. respiratory structures.
Moist cuticle : Earthworm : : Lung : _____
(1) Aquatic insects
(2) Aquatic mammals
(3) Fishes
(4) Aquatic molluscs

137. The conducting part of human respiratory system does **not** facilitate
(1) Humidification of inhaled air
(2) Temperature adjustment of inhaled air
(3) Clearing of foreign particles from the inhaled air
(4) Exchange of gases between alveolar air and blood

138. Select the **correct** statement with reference to normal breathing in humans.
(1) Normal inspiration is an active process whereas normal expiration is a passive process.
(2) Inspiration is a passive process whereas expiration is an active process.
(3) Both normal inspiration and expiration are active processes.
(4) Both inspiration and expiration are passive processes.

139. Read the following statements.
Statement A: Humans can forcefully expire with the help of additional muscles in abdomen.
Statement B: Humans have the ability to increase the strength of inspiration and expiration.
Choose the correct option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

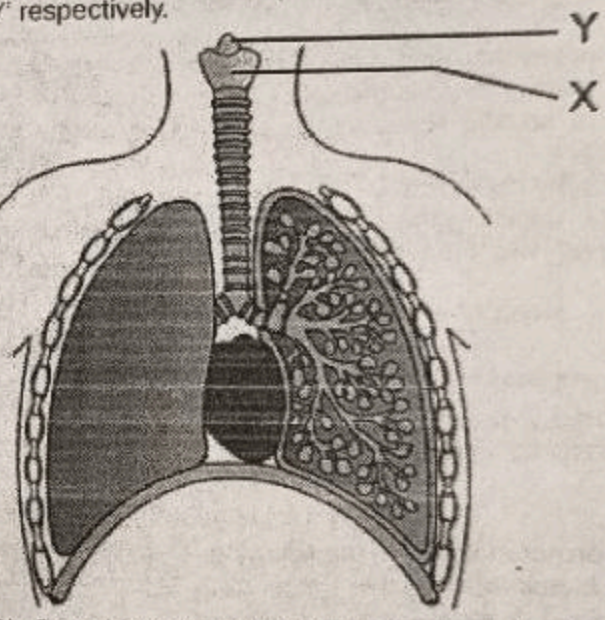
140. At the level of which vertebra does trachea divide into right and left primary bronchi?
(1) 4th thoracic vertebra
(2) 5th lumbar vertebra
(3) 5th thoracic vertebra
(4) 5th cervical vertebra

141. Ciliated epithelium lining the bronchioles helps in
(1) Exchange of gases
(2) Reabsorption of water
(3) Filtration of blood
(4) Movement of mucus and trapped particles in a specific direction through respiratory tract

142. Animals which have a network of tracheal tubes to transport atmospheric air within the body are
(1) Flatworms
(2) Insects
(3) Coelenterates
(4) Sponges

143. The process occurring within the cells, where oxygen is utilised by the organisms during metabolic reactions for production of ATP resulting in release of CO_2, is known as
(1) Pulmonary ventilation
(2) Alveolar ventilation
(3) Branchial respiration
(4) Cellular respiration

144. Identify the structures marked as 'X' and 'Y' in the figure given below and select the correct option regarding 'X' and 'Y' respectively.



(1) Sound box; Made up of non-elastic cartilaginous flap
(2) Thick elastic cartilaginous flap; Help in easy passage of food into trachea
(3) Sound box; Prevent entry of food into wind pipe
(4) Prevent entry of food into pharynx; Made up of non-elastic cartilage

145. In a dsDNA molecule, guanine compulsorily base pairs with
(1) Adenine
(2) Thymine
(3) Cytosine
(4) Uracil

146. Lungs have a large number of alveoli primarily for
(1) Having spongy texture and proper shape
(2) More surface area for diffusion of gases
(3) More space for increasing volume of inspired air
(4) More nerve supply

147. Structures which are **not** supported by incomplete cartilaginous rings are
(1) Tracheae
(2) Primary bronchi
(3) Initial bronchi
(4) Terminal bronchioles

horacic chamber is ventrally, dorsally and laterally formed, respectively, by
(1) Vertebral column, ribs, and sternum
(2) Ribs, sternum, and diaphragm
(3) Sternum, vertebral column, and ribs
(4) Diaphragm, sternum, and ribs

148. The positional information regarding sequence of amino acids in a protein is represented by its
(1) Quaternary structure (2) Secondary structure
(3) 3-D structure (4) Primary structure

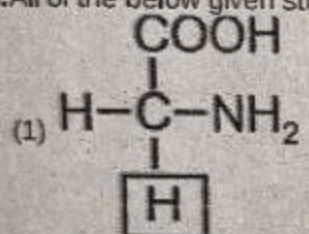
150. The simplest amino acid with lowest molecular weight and no asymmetrical carbon atom is
(1) Methionine (2) Alanine
(3) Glycine (4) Tryptophan

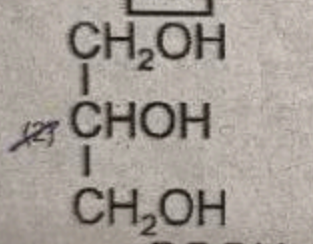
151. Which of the following is **not** an aromatic amino acid?
(1) Tyrosine (2) Phenylalanine
(3) Tryptophan (4) Alanine

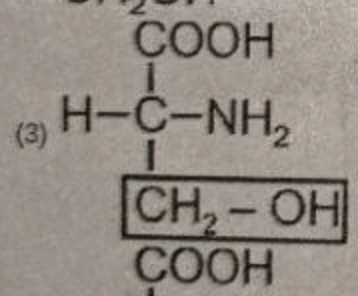
152. The **correct** representation of palmitic acid is
(1) $CH_3 - (CH_2)_{16} - COOH$
(2) $CH_3 - (CH_2)_{14} - COOH$
(3) $CH_3 - (CH_2)_7 CH = CH (CH_2)_7 COOH$
(4) $CH_3 - (CH_2)_{18} - COOH$

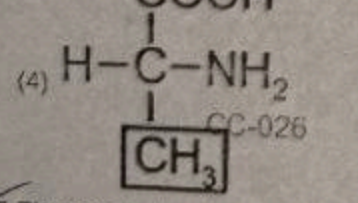
153. Cholesterol is an example of
(1) Phospholipid (2) Simple lipid
(3) Glycolipid (4) Derived lipid

154. All of the below given structures are of amino acids, **except**

(1) 

(2) 

(3) 

(4) 

155. The most abundant protein present in the animal world is 'P'.
Select the **correct** option for 'P'.
(1) Also found in the blood
(2) Arranged in an irregular pattern in dense connective tissue that is present beneath the skin
(3) Not stabilized by hydrogen bonds
(4) A storage homopolymer in plants

156. In a nucleotide, an ester bond is present between the
(1) Sulphate and carboxyl group of sugar
(2) Phosphate and hydroxyl group of sugar
(3) Carbonate and carboxyl group of sugar
(4) Carboxyl and phosphate group of sugar

157. Coenzyme nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin
(1) Thiamine (2) Niacin
(3) Pyridoxine (4) Tocopherol

158. All of the following bonds are likely to be involved in stabilizing the three-dimensional folding of most proteins, **except**
(1) Hydrogen bonds (2) Covalent bonds
(3) Ionic bonds (4) Phosphodiester bonds

159. Read the given statements and select the correct option.
Statement A: The type of co-factor that interacts with peroxidase is actually an organic compound which is tightly bound to the apoenzyme.
Statement B: Catalase and carboxypeptidase possess same type of inorganic co-factor like haem.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

160. In a hypothetical scenario, if 'E' represents enzyme 'S' represents substrate, 'P' represents product, 'ES' represents enzyme-substrate complex and 'EP' represents enzyme-product complex then, which of the following is the correct representation of an enzyme catalysed biochemical reaction?
(1) $E + S \rightarrow EP \rightarrow ES \rightarrow E + P$
(2) $E + P \rightarrow EP \rightarrow ES \rightarrow E + S$
(3) $E + ES \rightarrow EP \rightarrow ES \rightarrow S + P$
(4) $E + S \rightarrow ES \rightarrow EP \rightarrow E + P$

161. Select the class of enzyme involved in the following reaction.
$S - G + S' \rightarrow S + S' - G$
(1) Ligases (2) Oxidoreductases
(3) Hydrolases (4) Transferases

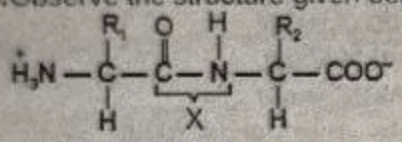
162. The components of a lecithin molecule are all of the following, **except**
(1) Nitrogen (2) Glycerol
(3) Fatty acids (4) Cholesterol

163. Michaelis constant (K_m) refers to substrate concentration at which the reaction velocity reaches ____ of its maximum velocity.
Choose the option that fills the blank **correctly.**
(1) Three-fourth (2) One-third
(3) One-fourth (4) Half

164. How many statements given below is/are **incorrect**?
(a) Enzymes are divided into 6 classes, each with 4-13 subclasses and named accordingly by a 4-digit number.
(b) Ligases catalyse the joining of C – O, C – S, C – N, P – O bonds, *etc.*
(c) Hydrolases catalyse the hydrolysis of ester, ether, peptide and glycosidic bonds.
(d) Catalytic activity is lost permanently when the enzymes are exposed to low temperature.
Choose the correct option.
(1) Two (2) Three
(3) Four (4) One

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165. Observe the structure given below.



X represents a/an
(1) Ionic linkage (2) Peptide bond
(3) Glycosidic bond (4) Ester bond

166. The enzyme which catalyses breaking of bonds between the glycerol and fatty acids in a molecule of triglyceride has enzyme commission number that starts with
(1) One (2) Two
(3) Three (4) Four

167. The maximum volume of air a person can breathe in after a forced expiration is about
(1) 1100 mL (2) 8000 mL
(3) 500 mL (4) 4500 mL

168. Which among the following options represents a pair of proteins that plays a role in transport of oxygen and glucose in human body, respectively?
(1) Insulin, Haemoglobin (2) Insulin, GLUT-4
(3) Haemoglobin, GLUT-4 (4) Collagen, Trypsin

169. There are _____ types of amino acids which occur in proteins and ___ types of nucleotides are present in RNA. Choose the **correct** option to fill in the respective blanks.
(1) 20, 4 (2) 20, 8
(3) 10, 5 (4) 10, 4

170. The sugar present in DNA is a
(1) Hexose sugar
(2) Pentose sugar in which oxygen is present at 2nd carbon
(3) Monosaccharide
(4) Dissaccharide

171. Adult human haemoglobin is made up of
(1) 2α and 2γ subunits (2) 2β and 2γ subunits
(3) 2α and 2β subunits (4) 3α and 1γ subunits

172. Match the items given in column I to the items given in column II and select the **correct** option.

Column I	Column II
a. FRC	(i) ERV + RV
b. VC	(ii) RV + ERV + IRV + TV
c. IC	(iii) TV + IRV
d. TLC	(iv) ERV + TV + IRV

(1) a(i), b(iv), c(iii), d(ii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(i), b(iv), c(ii), d(iii) (4) a(i), b(ii), c(iv), d(iii)

173. Cytidine is a/an
(1) Nucleoside (2) Nucleotide
(3) Essential amino acid (4) Non-essential amino acid

174. Based on the number of amino and carboxyl groups, valine and lysine are
(1) Acidic and basic amino acids, respectively
(2) Neutral amino acids
(3) Neutral and basic amino acids, respectively
(4) Basic and acidic amino acids, respectively

175. Which among the following is observed during the process of normal inspiration in humans?
(1) Flattening of diaphragm
(2) Contraction of abdominal muscles
(3) Decrease in the volume of thoracic chamber
(4) Relaxation of the external inter-costal muscles

176. Most abundant protein in whole of the biosphere is
(1) RuBisCO (2) Collagen
(3) Albumin (4) Globulin

177. In humans, lungs are covered by a _____ layered pleura.
Fill the blank with a suitable option.
(1) Single (2) Double
(3) Triple (4) Tetra

178. The volume of air that **cannot** be estimated by using a spirometer is
(1) Tidal volume
(2) Inspiratory reserve volume
(3) Residual volume
(4) Expiratory reserve volume

179. The rate of breathing in a healthy adult human is
(1) 20-25 times/minute (2) 12-16 times/minute
(3) 12-16 times/second (4) 40-45 times/hour

180. Structure in lungs which is directly involved in exchange of O_2 and CO_2 between air and blood in humans is
(1) Bronchi (2) Trachea
(3) Alveoli (4) Nasal chambers

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$t = \frac{45}{2} = 15$

